

Networks Assignment - 2

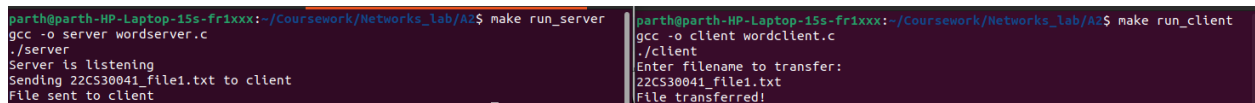
- Parth Shashin Patil
- 22CS30041

Link to .pcap file (g-drive) -

<https://drive.google.com/file/d/1D-C1VmfwQ1m-aEOM3uV6uMR-sVzY3xC/view?usp=sharing>

1) Packet Capture

Terminal Screenshot -



```
parth@parth-HP-Laptop-15s-fr1xxx:~/Coursework/Networks_lab/A2$ make run_server
gcc -o server wordserver.c
./server
Server is listening
Sending 22CS30041 file1.txt to client
File sent to client

parth@parth-HP-Laptop-15s-fr1xxx:~/Coursework/Networks_lab/A2$ make run_client
gcc -o client wordclient.c
./client
Enter filename to transfer:
22CS30041_file1.txt
File transferred!
```

Wireshark Screenshot -

1	0.000000000	127.0.0.1	127.0.0.1	UDP	62 60052 → 5000 Len=20
2	0.000195046	127.0.0.1	127.0.0.1	UDP	48 5000 → 60052 Len=6
3	0.000356346	127.0.0.1	127.0.0.1	UDP	47 60052 → 5000 Len=5
4	0.000383278	127.0.0.1	127.0.0.1	UDP	47 5000 → 60052 Len=5
5	0.000415683	127.0.0.1	127.0.0.1	UDP	47 60052 → 5000 Len=5
6	0.000435139	127.0.0.1	127.0.0.1	UDP	45 5000 → 60052 Len=3
7	0.000544858	127.0.0.1	127.0.0.1	UDP	47 60052 → 5000 Len=5
8	0.000567260	127.0.0.1	127.0.0.1	UDP	44 5000 → 60052 Len=2
9	0.000591802	127.0.0.1	127.0.0.1	UDP	47 60052 → 5000 Len=5
10	0.000604751	127.0.0.1	127.0.0.1	UDP	47 5000 → 60052 Len=5
11	0.000630766	127.0.0.1	127.0.0.1	UDP	47 60052 → 5000 Len=5
12	0.000645861	127.0.0.1	127.0.0.1	UDP	49 5000 → 60052 Len=7

2) The protocol used in the given program is **User Datagram Protocol (UDP)**.

3) Client to Server message -

SOURCE IP - 127.0.0.1 (localhost)
SOURCE PORT - 60052
DESTINATION IP - 127.0.0.1 (localhost)
DESTINATION PORT - 5000

Server to Client message -

SOURCE IP - 127.0.0.1 (localhost)
SOURCE PORT - 5000

DESTINATION PORT - 60052

5) The size of the server's response for HELLO and the first word (*WORD1*) is **48 bytes** (6 bytes excluding headers) and **47 bytes** (5 bytes excluding headers) respectively.

```

Frame 4: 47 bytes on wire (376 bits), 47 bytes captured (376 bits) on interface 0
Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00)
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
User Datagram Protocol, Src Port: 5000, Dst Port: 60052
Data (5 bytes)
  Data: 7468697300
  [Length: 5]

```

8) Average packet size - **48 bytes** (6 bytes excluding headers).